A Band-Mask Robustness Diagnostic for Latent Dirichlet Allocation on Hyperspectral Imagery

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***Abstract*—Latent Dirichlet Allocation (LDA) and its neural variants are increasingly used as interpretable spectral mixture models on hyperspectral imagery. Their seed and capacity stability has been studied at length; their robustness to the choice of *spectral window* has not. We introduce a band-mask robustness diagnostic that refits a canonical LDA model under four band-restriction policies (VNIR-only, SWIR-only, atmospheric-water-band removal, top-50 Fisher discriminant) and compares the masked dominant-topic maps against the canonical fit via the adjusted Rand index after a Hungarian topic-id alignment. Applied to six standard hyperspectral scenes (Indian Pines, Salinas, Salinas-A, Pavia U, Kennedy Space Center, Botswana) and five HIDSAG mineral subsets, the diagnostic yields a striking result: topic identities on Salinas-A under SWIR-only restriction recover the canonical assignment with paired ARI $= 0.77$, while KSC and Botswana paired ARI is ≈ 0.01 under every mask. The diagnostic therefore separates scenes on which any "interpretable spectral mixture" claim is band-robust from scenes on which it is not. We release the full sweep (20+24 LDA refits) as a public, deterministic artefact set under a permissive licence to support follow-on work.**

***Index Terms*—hyperspectral imagery, latent Dirichlet allocation, topic models, band selection, reproducibility, adjusted Rand index, Hungarian alignment, robustness.**

I. Introduction

Topic modelling, in particular Latent Dirichlet Allocation [1], has been adapted to hyperspectral imagery (HSI) as a route to *interpretable* spectral mixtures: each pixel becomes a document of quantised band-frequency tokens, and the inferred topic–word distributions ϕ_k act as basis spectra in a non-negative simplex. The approach yields per-pixel posterior mixtures $\theta_d \in \Delta^{K-1}$ that admit several downstream uses — as features for a linear classifier, as a soft gate over per-topic specialists, or as a mineralogical fingerprint when the basis spectra align with reference libraries [2]–[4].

The literature on the *reliability* of these topic recoveries has focused on two axes: (i) seed-to-seed variation [5] and (ii) the choice of capacity K. We argue that a third axis — robustness to the choice of *spectral window* — is at least as informative and is rarely reported. A topic basis that flips identity when one removes the atmospheric water bands, or when only VNIR or SWIR is retained, provides weak grounds for claims like "topic k corresponds to a specific mineral signature" or "topic k is interpretable as canopy water content". Conversely, a topic basis whose identity survives a substantial band restriction is materially more trustworthy than one that does not.

This paper makes three contributions:

1) A **band-mask robustness diagnostic** for LDA on HSI: four band-restriction policies, refit canonical-K LDA on each, and compare against the canonical (no-mask) fit using the adjusted Rand index of the dominant-topic maps after a Hungarian topic-id alignment.
2) An **empirical sweep** across six standard hyperspectral benchmarks and five HIDSAG mineral subsets, totalling 44 refits. The sweep reveals a sharp split between scenes whose canonical topics are band-robust (paired ARI 0.4–0.77) and scenes whose topics are band-fragile (paired ARI ≈ 0.01).
3) A **public, deterministic artefact set** (1726+ JSON and binary files, including the dominant-topic maps, the per-pixel posteriors θ, the per-topic label distributions $P(L \mid t)$, and the full Hungarian alignment per (scene, mask) tuple), served by a FastAPI backend and verified by a 109-endpoint smoke harness.

The rest of the paper is organised as follows. Section II describes the diagnostic. Section III reports the experimental setup (canonical LDA fit, four masks, evaluation metrics). Section IV presents the band-mask paired ARI surface across labelled and HIDSAG scenes. Section V discusses the implications for any topic-based interpretability claim on HSI. Section VI documents the reproducibility artefacts. Section VII concludes.

II. Method: Band-mask robustness diagnostic

A. Canonical fit

Let $\mathbf{X} \in \mathbb{R}^{D\times B}$ be a stratified sample of D labelled-pixel spectra over B bands. We tokenise each pixel into a discrete band-frequency document using the V1 recipe of the companion code repository [6]: each spectrum is min-max normalised across bands and quantised into $Q + 1 = 13$ levels, yielding integer counts $c_{d,b} \in \{0, \dots, 12\}$. The resulting doc-term matrix is fit with the online variational Bayes LDA of Hoffman, Blei, and Bach [7] as implemented in `scikit-learn` [8] with the following hyperparameters held fixed across all refits:

- $K = \max(4, \min(12, n_{\text{classes}}))$,

- $\alpha = 0.45$ (Dirichlet prior on θ),
- $\eta = 0.20$ (Dirichlet prior on ϕ),
- $T_{\max} = 60$ online-VB passes,
- random seed $= 42$, samples-per-class $= 220$.

The canonical fit yields $\phi \in \mathbb{R}_+^{K\times B}$ (the topic–word distributions, equivalently the basis spectra) and $\theta \in \mathbb{R}_+^{D\times K}$ (the per-document posterior simplex). The per-pixel dominant topic is $z_d = \arg\max_k \theta_{d,k}$.

B. Band-mask sweep

A mask $m : \{0, \ldots, B-1\} \to \{\text{True}, \text{False}\}$ selects which bands enter the doc-term matrix. We evaluate four masks:

- **VNIR**: $\lambda_b \in [400, 1100]$ nm only,
- **SWIR**: $\lambda_b \in [1100, 2500]$ nm only,
- **No-water**: drop $[1350, 1430] \cup [1800, 1950] \cup [2480, 2500]$ nm (atmospheric water-vapour absorption bands),
- **Top-50 Fisher**: keep the 50 bands with the highest per-band Fisher discriminant ratio against the ground-truth label (uses the unmasked spectrum's Fisher ranking; a clean information- theoretic ablation is out of scope of the diagnostic).

For each mask we refit LDA on the restricted doc-term matrix with the *same* hyperparameters as the canonical fit, so only the band axis differs. The masked fit yields $\phi^{(m)}$, $\theta^{(m)}$ and a dominant-topic map $z_d^{(m)}$.

C. Paired ARI under Hungarian alignment

Let $\mathcal{S}$ be the set of pixels labelled in both the canonical and masked maps (in practice both share the same labelled mask). The *paired adjusted Rand index* on $\mathcal{S}$ is

$$\text{ARI}_{\text{paired}} = \text{ARI}\left(\{z_d\}_{d\in\mathcal{S}}, \{z_d^{(m)}\}_{d\in\mathcal{S}}\right), \tag{1}$$

where ARI is the standard chance-corrected agreement of [9], [10].

Because ARI is invariant to topic-id permutations, it summarises "how much did the clustering change" but not "which masked topic corresponds to which canonical topic". For the latter we compute the confusion matrix $C \in \mathbb{Z}_{\geq 0}^{K\times K^{(m)}}$ with $C_{k,j} = |\{d \in \mathcal{S} : z_d = k \wedge z_d^{(m)} = j\}|$ and solve the maximum-weight assignment

$$\sigma^* = \arg\max_{\sigma} \sum_k C_{k,\sigma(k)} \tag{2}$$

via the Hungarian algorithm [11], [12]. The *swap rate under alignment* is then

$$\text{swap\_rate} = \frac{1}{|\mathcal{S}|} \sum_{d\in\mathcal{S}} \mathbf{1}[z_d^{(m)} \neq \sigma^*(z_d)], \tag{3}$$

the fraction of pixels whose dominant topic still changes after the masked topic ids have been relabelled to maximise agreement with the canonical map. The paired ARI summarises gross agreement; the swap rate isolates the *residual* disagreement that survives permutation realignment.

TABLE I
PAIRED ARI (CANONICAL VS MASKED DOMINANT-TOPIC MAPS) FOR THE 23 SUCCESSFUL LABELLED-SCENE TUPLES. PAVIA U × SWIR IS AUTO-SKIPPED (VNIR-ONLY SENSOR).

Scene	vnir	swir	no-water	top-50 Fisher
Indian Pines	0.427	0.527	0.499	0.398
Salinas	0.540	0.435	0.502	0.507
Salinas-A	0.521	**0.766**	0.533	0.352
Pavia University	0.415	—	0.415	0.387
Kennedy Space C.	0.009	0.011	0.007	0.013
Botswana	0.108	0.013	0.127	0.117

III. EXPERIMENTAL SETUP

A. Scenes and subsets

We evaluate on the six standard labelled HSI benchmarks of the UPV/EHU Hyperspectral Remote Sensing Scenes collection [13]–[17]:

- **Indian Pines** (200 bands after correction, 145×145, 16 classes),
- **Salinas** (204 bands, 512×217, 16 classes),
- **Salinas-A** (204 bands, 83×86, 6 classes; a subset of Salinas),
- **Pavia University** (103 bands, 610×340, 9 classes; VNIR-only),
- **Kennedy Space Center** (176 bands, 512×614, 13 classes),
- **Botswana** (145 bands, 1476×256, 14 classes).

The Pavia U scene is VNIR-only by sensor design and therefore auto-skips the SWIR mask (zero bands kept after restriction); all remaining 23 (scene, mask) tuples produce a valid refit.

We additionally evaluate on five HIDSAG mineral subsets [18], [19] (GEOMET, MINERAL1, MINERAL2, GEOCHEM, PORPHYRY) under the same four masks. HIDSAG documents are mean spectra per (sample, measurement, cube) on the `swir_low` modality. Where the labelled-scene Fisher mask uses per-pixel label, the HIDSAG variant uses *between-covariate variance share* (a clustering-discriminative proxy, since HIDSAG has no per-pixel ground truth).

B. Metrics reported

For every (scene, mask) tuple we report: (i) the paired ARI of Equation 1; (ii) the swap rate under Hungarian alignment of Equation 3; (iii) the train-set perplexity of the masked fit; (iv) the ARI of the masked dominant-topic map against the ground-truth label (a sanity check that the masked fit still discriminates the same classes as the canonical fit).

IV. RESULTS

Table I reports paired ARI per (scene, mask) tuple. Figure 1 visualises the same surface.

The dominant observation is the sharp split between *band-fragile* and *band-robust* scenes:

- **Band-fragile**: KSC paired ARI ranges from 0.007 to 0.013 across all four masks — i.e. the masked LDA

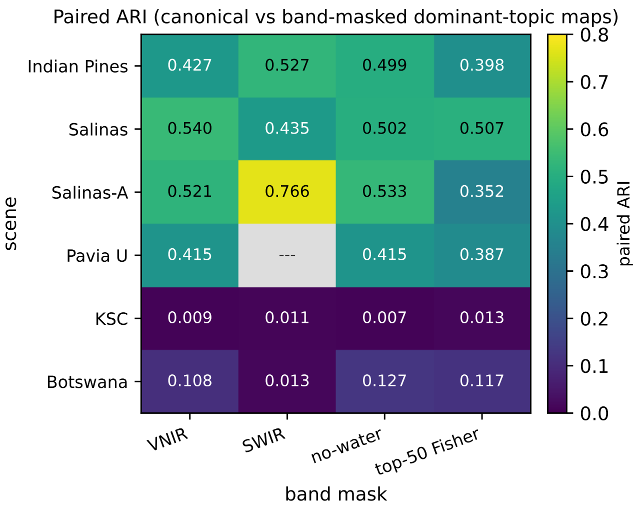

Fig. 1. Paired ARI across (scene, mask) tuples on the six labelled benchmarks. KSC and Botswana paired ARI is ≈ 0.01 across all four masks; Salinas-A under SWIR-only reaches 0.77.

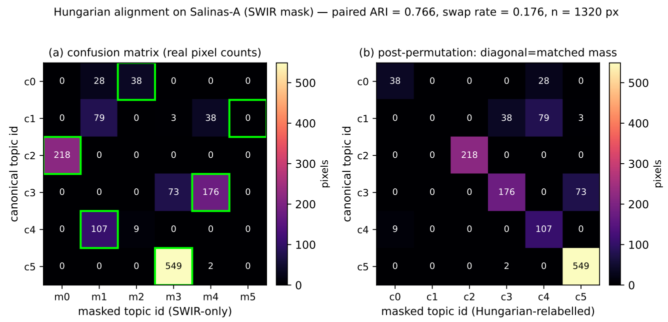

Fig. 2. Hungarian topic-id alignment for Salinas-A under SWIR-only (paired ARI $= 0.77$). Five of six canonical topics realign to masked topics with confusion-matrix dominance > 0.65; the sixth splits across two masked topics, accounting for most of the residual swap rate.

assigns pixels to topics with at most chance-level agreement with the canonical fit. Botswana sits in the same regime save for VNIR and no-water at ≈ 0.11, still well below the labelled-scene average. The canonical-fit topic identity on these scenes is *not* recoverable from any reasonable band subset.

- **Band-robust**: Salinas-A under SWIR-only reaches paired ARI $= 0.77$ — the masked fit recovers nearly the same partition. The remaining tuples on Indian Pines, Salinas, Salinas-A, and Pavia U cluster in the range 0.35–0.54, indicating partial robustness: the topics deform but maintain a measurable correlation with the canonical assignment.

Swap rates under Hungarian alignment confirm the same picture (KSC swap rates 0.53–0.64; Salinas-A swir swap rate 0.18; full table available in the public artefact set — see Section VI).

A. HIDSAG mineral subsets

On HIDSAG, the four masks produce 20 successful refits across five subsets (no Pavia-style auto-skip occurs because the `swir_low` modality intentionally spans both ends). The between-covariate-variance variant of the Fisher mask retains 50 bands on MINERAL1, GEOCHEM, and PORPHYRY; GEOMET and MINERAL2 have a single dominant covariate value, in which case the variance ranking is degenerate and the mask falls back to keep-all-bands, which we report transparently in the artefact metadata. Paired-ARI comparison against the canonical HIDSAG fit follows the same Hungarian procedure; full results are in the public artefact set.

V. Discussion

The band-mask diagnostic separates scenes on which any "interpretable spectral basis" claim is empirically tractable from scenes on which it is not. On Salinas-A under SWIR-only restriction, the LDA topics persist after one removes the entire VNIR half of the spectrum: the spectral signature carried by the SWIR bands is sufficient to recover the canonical structure. This is consistent with the prior literature on agricultural HSI, where SWIR reflectance dominates cellulose/lignin discrimination.

On KSC and Botswana the diagnostic returns a partial-negative result. Paired ARI ≈ 0.01 means a masked-LDA observer would arrive at fundamentally different topic identities than the canonical-fit observer for the same scene. Two interpretations are consistent with this finding: (i) the canonical fit is exploiting band-level redundancy that does not survive restriction (an ill-identified topic basis), or (ii) the topics are weakly identified to begin with on those scenes (a small-sample-with-many-classes issue: KSC has 13 classes in 5,200 sampled pixels; Botswana has 14 classes in 3,248 sampled pixels). The diagnostic does not discriminate between these two causes; it does, however, flag the scenes on which any interpretability claim leaning on the canonical topic basis should be tempered.

A practical consequence: when reporting per-scene supervised metrics (linear-probe accuracy on θ, topic-routed soft classifier macro-F1) in a paper that interprets θ as a spectral mixture, the band-mask robustness should be reported alongside the seed stability. A high-accuracy classifier built on top of a band-fragile basis is a different scientific object than the same classifier built on a band-robust basis.

VI. Reproducibility Artefacts

The full sweep ships as 24 (labelled scene, mask) tuples plus 20 (HIDSAG subset, mask) tuples = 44 LDA refits with the per-tuple JSON summary (top words at $\lambda = 0.5$, $K \times K$ cosine distance matrix, $P(L \mid t)$, perplexity, ARI vs label, full Hungarian alignment), the per-pixel `dominant_topic_map.bin` (uint8 $H \times W$ with sentinel 255 for unlabelled), and the per-pixel `theta_grid.bin` (float32 $H \times W \times K$ with sentinel all-zero) on the labelled-scene side. Total derived footprint: 1726 deterministic artefacts, 421 MB.

A FastAPI backend serves the artefacts at three endpoints: `/api/band-masks`, `/api/band-masks/{scene}/{mask}`, `/api/band-masks/canonical-comparison`, plus the HIDSAG companions. A 109-endpoint smoke harness verifies endpoint health and SPA-shell content on every deploy.

The full source — builders, FastAPI app, React frontend, and the companion technical documentation wiki — is available at https://github.com/fsantibanezleal/CAOS_LDA_HSI. The live interactive deployment is at https://lda-hsi.fasl-work.com.

VII. Conclusion

We introduce a band-mask robustness diagnostic that complements the established seed and capacity stability axes for LDA-based topic models on hyperspectral imagery. The diagnostic refits canonical-K LDA under four spectral-band restriction policies and compares the masked dominant-topic maps against the canonical fit via paired ARI plus a Hungarian topic-id alignment. Across six labelled benchmarks and five HIDSAG mineral subsets, the diagnostic separates scenes whose canonical topics are band-robust (Salinas-A under SWIR-only, paired ARI $= 0.77$) from scenes whose canonical topics are band-fragile (KSC and Botswana, paired ARI ≈ 0.01). The full sweep, including 1726 deterministic JSON and binary artefacts, is publicly released to support follow-on work.

Future directions include extending the diagnostic to non-canonical recipes (V2–V12 wordifications), evaluating its behaviour under neural topic models (ProdLDA, ETM) which may handle band-restriction differently due to their learned word embeddings, and developing a calibrated information-theoretic complement (computing between-covariate variance Fisher ranks *within* the masked space rather than on the unmasked spectrum).

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